

Simple Random Sample

SurvMeth/Surv 625: Applied Sampling

Yajuan Si

University of Michigan, Ann Arbor

1/15/25

Simple random sample (SRS)

- Implementation: Selection
- Inference: Analysis
- Projection: Future collection
- Sampling frame

Implementation

- SRS is the most basic form of probability sampling and provides the basis for the more complicated forms.
- Select a random sample of size n from a population of N
 - ① SRS with replacement (SRSWR): the same unit can be included more than once in the sample, with the equal selection probability $\pi_i = n/N$
 - ② SRS without replacement (SRS): all units in the sample are distinct with the equal selection probability $\pi_i = n/N$
- Uniform random number generator
 - R functions: `sample()`, `sampling::srswor()`, `sampling::srswr()`, etc.
- Reproducible: `set.seed()`

Inference: Population mean

- Suppose we collect the systolic blood pressure (SBP) measurements of the five individuals selected via SRSWOR from a population of 20 and would like to estimate the population average SBP value: 110, 125, 145, 90, 135
- The finite population correction factor
$$fpc = 1 - f = 1 - n/N = 1 - 5/20 = 15/20$$
- The sample total $t = 110 + 125 + 145 + 90 + 135 = 605$
- The sample mean $\bar{y} = t/n = 605/5 = 121$
- The element variance estimate $s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2 = 467.5$
- The standard deviation $s = \sqrt{s^2} = 21.622$

Inference: Population mean cont.

- The sampling variance estimate $var(\bar{y}) = (1 - f) \frac{s^2}{n} = 70.125$
- The standard error $se(\bar{y}) = \sqrt{var(\bar{y})} = 8.374$
- The 95% confidence interval

$$\bar{y} \pm t_{1-\alpha/2, n-1} * se(\bar{y}) = 121 \pm t_{0.975, t} * 8.374,$$

where the critical t-score $t_{0.975, t} = 2.776$ (NOTE: not 1.96)

Inference: Population total

- Population total estimate: $\hat{t} = N\bar{y} = 20 * 121$
- The sampling variance of the population total estimate:

$$var(\hat{t}) = N^2 var(\bar{y}) = N^2(1 - f) \frac{s^2}{n} = 20^2 * 70.125$$

The standard error: $se(\hat{t}) = \sqrt{var(\hat{t})} = N\sqrt{var(\bar{y})} = 20 * 8.374$

- The $1 - \alpha$ confidence interval: $\hat{t} \pm t_{1-\alpha/2, n-1} se(\hat{t})$

Inference: Population proportion

- Now we are interested in estimating the proportion of individuals with hypertension, and hypertension indicator values are: 0, 0, 0, 1, 0
- The proportion estimate: $p = \bar{y} = 1/5$
- The theoretical sampling variance (UNKNOWN):

$$Var(p) = (1 - f) \frac{S^2}{n} = \frac{1-f}{n} \frac{N}{N-1} P(1 - P)$$

- The sampling variance estimate:

$$var(p) = \frac{1 - f}{n} \frac{n}{n - 1} p(1 - p) = (1 - 5/20)/4 * 1/5 * (1 - 1/5)$$

- The sampling error: $se(p) = \sqrt{var(p)}$
- The $1 - \alpha$ confidence interval: $p \pm t_{1-\alpha/2, n-1} se(p)$

R Code

```
library(survey)
library(sampling)
library(SDAResources)
library(tidyverse)
data(agpop)
n <- length(agsrs$acres92)
ybar <- mean(agsrs$acres92, na.rm = T)
ybar
```

```
[1] 297897
```

```
hatvybar <- (1 - n/3078) * var(agsrs$acres92, na.rm = T) / n
seybar <- sqrt(hatvybar); seybar
```

```
[1] 18898.43
```

```
# Calculate confidence interval by direct formula using t distribution
Mean_CI <- c(ybar - qt(.975, n-1)*seybar, ybar + qt(.975, n-1)*seybar)
names(Mean_CI) <- c("lower", "upper"); Mean_CI
```

```
      lower      upper
260706.3 335087.8
```

```
# To obtain estimates for the population total,
# multiply each of ybar, seybar, and Mean_CI by N = 3078
seybar*3078; Mean_CI*3078
```

```
[1] 58169381
```

```
      lower      upper
802453859 1031400361
```

```
# Calculate coefficient of variation of mean
seybar/ybar
```


Projection

- When designing a new survey data collection, investigators often focus on one or two key variables of interest, decide the amount of sampling error, and must balance the estimation precision with survey costs.
- ① Specify the tolerable error: What is expected of the sample, and how much precision do I need? The **desired precision** is often expressed as $P(|\bar{y} - \bar{y}_u| \leq e) = 1 - \alpha$, where e is called the *margin of error*, as one-half of the width of a 95% CI. Sometimes you like to achieve a desired relative precision, such as a desired CV.
- ② Find an equation: Relating the sample size n to the desired precision.
- ③ Estimate any unknown quantities and solve for n .
- ④ Adjust expectations and re-calculate.

Example of sample size projection

- Desired precision

- ① $var(\bar{y}) = \hat{V} = 2.5;$

- ② Requested 95% CI: $(L, U) = (10, 20)$, we have
 $e = (U - L)/2 = z_{0.975} * \hat{V}$ and then $\hat{V} = 2.5$

- ③ Desired CV: $cv(\bar{y}) = \frac{\hat{V}}{\bar{y}} = 0.5$. If $\bar{y} = 5$, then $\hat{V} = 2.5$

- Find an equation

- ① SRSWR: $n_0 = \frac{\hat{S}^2}{\hat{V}^2} = \frac{z_{0.975}^2 \hat{S}^2}{e^2}$

- ② SRS: $n = \frac{n_0}{1 + \frac{n_0}{N}}$

- Solve for n : If $\hat{S}^2 = 6250$, $\hat{V} = 2.5$, and $N = 100000$, then
 $n_0 = 1000$ and $n = 991$.

Design effects

- **Design effect:** Ratio of the variance under a new design to SRS variance with the same sample size
- Depend on the estimated quantity (e.g., mean, regression coefficient)
- Depend on the examined variable
- Often used as a comparison to SRS when evaluating complex sample survey designs
- Inflate the projected SRS sample size by the design effect for complex sample survey sample size projection

Sampling frame

- Frame: Set of materials used to designate a sample of units
- Rule links frame elements to population elements
- Accurate and up-to-date frames located in one location preferred
- Numbered, computerized lists are best
- The population and the list/frame may not match up

Frame problems: Noncoverage

- Some population elements are not on the frame
 - Have zero chance of selection
- Potential solutions
 - 1 Use supplemental frames that cover noncovered elements
 - 2 Use noncoverage weighting adjustments (will be discussed later)

Frame problems: Blanks

- Frame elements do not have corresponding population elements
- Know frame element is blank after selection
 - Screening to find eligible list elements
- Potential solutions
 - 1 Reject blanks by adjusting the sampling rate and size
 - 2 Substitute with the next element on the listing

Frame problems: Duplicates

- Occur when a single population element is linked to two or more frame elements
- Potential solutions
 - ① If only a few readily identified, remove from the list before selection
 - Eliminating duplicates from the sample still leaves unequal probabilities of selection
 - ② Choose unique listing
 - First, last, largest, or randomly chosen frame listing
 - ③ Determine how many duplicates for a given selected element and weight

Frame problems: Clustering

- Occurs when more than one population element can be selected by a sample frame element
- Potential solutions
 - 1 Take all elements within selected clusters
 - The sample size varies with unequal-size clusters (discuss later)
 - Can adjust the sample size in advance
 - 2 Use cluster sampling (discuss later)
 - 3 Weighting adjustment

Objective respondent selection

- In the social sciences, sampling a single element from small clusters of unequal size occurs often
 - Sampling individuals from households
- Techniques for households:
 - Nearest birthday method
 - Objective Respondent Selection proposed by Kish

Within household selection: The Kish method

- Interviewers list eligible household members by gender and age
- Use selection table
- Selection tables “rotated” across households
- Maximum of four eligibles per household can be handled
 - Can be expanded to handle households with five/six eligibles

Respondent selection tables

Table A (1/4)	
If number of eligible subjects is	Select subject number
1	1
2	1
3	1
4	1

Table B (1/12)	
If number of eligible subjects is	Select subject number
1	1
2	1
3	1
4	2

Table C (1/6)	
If number of eligible subjects is	Select subject number
1	1
2	1
3	2
4	2

Table D (1/6)	
If number of eligible subjects is	Select subject number
1	1
2	2
3	2
4	3

Table E (1/12)	
If number of eligible subjects is	Select subject number
1	1
2	2
3	3
4	3

Table F (1/4)	
If number of eligible subjects is	Select subject number
1	1
2	2
3	3
4	4

Frame problems: Many-to-many matching

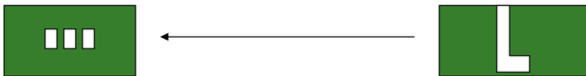
- Occurs when more than one population element can be selected by more than one frame element
- Potential solutions
 - Combinations of weighting and subsampling

Frame problems: Summary

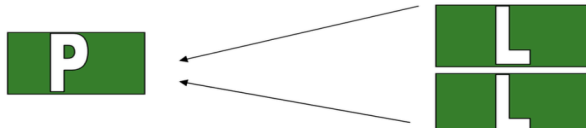
- Non-coverage



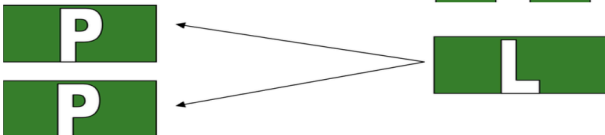
- Blanks



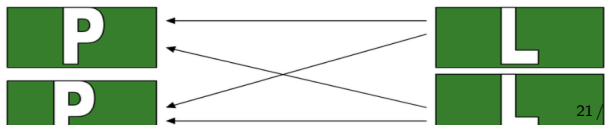
- Duplicates



- Clustering



- Many to many



Example: Address-based sample of families

- A survey about families and their use of credit to finance purchases of durable goods (for example, cars, household appliances) is to be given to a sample of families in a large metropolitan area.
- A random sample of $n = 5,000$ addresses is selected from the list of $N = 872,000$ “Delivery Sequence File” addresses for the metropolitan area purchased from a vendor of the United States Postal Service.
- The addresses are in order by zip code (approximately 15,000 addresses each), carrier route, and delivery sequence (the sequence through a carrier route followed to deliver mail).

Potential frame problems and solutions

- Noncoverage: families at addresses where the USPS does not deliver mail (for example, address uses a PO box only).
 - Remedy: adjust the estimates through weighting to compensate for non-coverage.
- Blanks: addresses that are not residential units, and do not contain families.
 - Remedy: skip the non-residential units, and increase the sample number of addresses slightly to account for blank listings.
- Clustering: addresses might be clusters of families (e.g., an apartment building with one address).
 - Remedy: Select all families at an address.

Summary: When to use SRS?

- SRS is the simplest of all probability sampling methods: Objective, randomized selection
- Every element has an equal chance of selection, called as *epsem*, or self-weighting
- However, the practical use of SRS is rare as it requires
 - ① A list of population observation units
 - ② Little extra information is available
 - ③ Analysis assuming independent observations and using SRS formulas
 - ④ High cost
- Even “bad” samples have an equal chance of being selected